Project Overview

This project is an Exploratory Data Analysis (EDA) of Kickstarter crowdfunding campaigns, built entirely in Power BI. The goal is to understand what factors drive project success, explore trends across categories and countries, and deliver an interactive three-page dashboard.

Objectives:

- Identify the key factors that distinguish successful from failed campaigns.
- Explore funding patterns across categories, countries, and time periods.
- Build an interactive Power BI dashboard with three focused report pages.

Tools:

- Power BI Desktop — Power Query for data preparation, DAX for measures and calculated columns, Report view for visualisations.

Data Source:

- Kaggle — Kickstarter Projects Dataset (kemical)
- Two CSV files provided: ks-projects-201612.csv and ks-projects-201801.csv.

Dataset Details

This dataset was obtained from the Kaggle platform and contains information about more than 300,000 Kickstarter projects. The data was originally collected from the Kickstarter crowdfunding platform and includes various attributes related to each project such as category, funding goal, pledged amount, and project status.

The dataset is widely used for data analysis and machine learning tasks, especially for predicting project success and understanding the factors that influence crowdfunding outcomes.

The dataset contains multiple features describing each Kickstarter project, including:-

- ID:
- Name:
- Category:
- main_category:
- Currency:
- Deadline:
- Launched:
- Goal:
- Pledged:
- state:
- backers:
- Country:
- usd_pledged:
- usd_pledged_real:
- usd_goal_real:

Note: Kickstarter is an American platform founded in New York in 2009. It remained exclusive to US residents for a long time before gradually opening up to other countries. This explains the significant bias towards US projects in the data.

Data Preparation (Power Query)

This phase involved importing the two available CSV files, cleaning and standardising their structure, combining them, and finally selecting the appropriate dataset for analysis. All cleaning steps were applied before merging (Early Transformation) to reduce the volume of data being processed at each stage.

Step 1: Import the Data

The dataset is provided in two separate CSV files:

- ks-projects-201612.csv — contains projects up to December 2016.
- ks-projects-201801.csv — contains projects up to January 2018.

1. Open Power BI Desktop.
2. Click Get Data → Text/CSV.
3. Select ks-projects-201612.csv.
4. In the preview window, click Transform Data (this opens Power Query Editor).
5. In Power Query, click New Source → Text/CSV.
6. Select ks-projects-201801.csv.

Step 2: Cleaning and Structuring the 2016 File

During the data type validation phase, it was observed that the file contained commas within the name field. The commas were interpreted as column delimiters instead of being treated as part of the field. This caused the data to be split incorrectly, shifting subsequent fields to the right.

As a result:

- The file was initially read as having 17 columns instead of the expected 13. Data that should have appeared in later columns (e.g., category, main_category, goal) were shifted to the right, creating four extra columns (Column1 to Column4) at the end of the file.

The following cleaning steps were applied:

- Column headers promoted — the first row of the CSV was promoted to become column names.
- Column names trimmed — any leading or trailing spaces were removed to avoid reference errors. Column headers must be identical to avoid errors during Append.
- Data types assigned — basic types were set (e.g., ID as integer, goal and pledged as numbers, state and category as text).
- Splitting date-time columns — both deadline and launched were stored as text in the format "YYYY-MM-DD HH:MM:SS". To work with dates only, each column was split into a date part and a time part. The time part was then removed, and the remaining date column was renamed back to the original column name.
- The four extra columns were removed, as they contained data that was no longer aligned with the intended structure.
- Rows where the shift had caused critical fields to be unrecoverable were removed. The number of such rows was small (~570 rows, less than 0.2% of total data), and their exclusion does not materially affect the analysis. This approach was chosen as the simplest solution.
- Source year added — a custom column dataYear was created with the value 2016 to keep track of the original file. This column is optional but useful for auditing.

After these steps, the 2016 file was ready for later combination with the 2018 file.

Step 3: Cleaning and Structuring the 2018 File

The 2018 file contained 15 columns, including two additional USD conversion columns that the 2016 file lacked:

- usd_pledged_real — pledge amount converted to USD using external exchange rates.
- usd_goal_real — goal amount converted to USD using external exchange rates.

The cleaning steps applied were similar to those for the 2016 file, with a few differences:

- Column headers promoted and trimmed — same as before.

- Data types assigned — all columns were given appropriate types; notably, deadline was correctly interpreted as a date, so no splitting was needed.
- Unix Epoch records removed — during data type validation, some records were found to have incorrect launch dates, specifically 01/01/1970. This date is a known default caused by Unix timestamp conversion errors or missing values. These records resulted in unrealistic project durations and were removed to ensure data quality and accurate analysis. The earliest valid launch date in the dataset was confirmed to be in 2009, which aligns with Kickstarter's founding year.
- Splitting launched — the launched column again contained a date and time, so it was split into date and time parts, and the time part was removed.
- Duration column added — a calculated column Duration was created using the formula: Duration = deadline − launched (result in days).
- Source year added — a dataYear column was created with the value 2018.

Step 4: Combining the Two Files

After both files were cleaned individually, they were combined into one table using the Append Queries feature in Power Query. This operation stacked the rows of the 2016 file on top of the rows of the 2018 file, creating a single unified table "AllData".

Step 5: Addressing Duplicate Records

Upon reviewing the appended data, it was observed that every project from the 2016 file also existed in the 2018 file (i.e., the 2016 file is a complete subset of the 2018 file). As a result, the appended table contained duplicate records for the same projects.

Two options were considered:

- Keep only unique projects — removing duplicates based on the ID column.
- Use only the 2018 file — since it already contains all available data and includes the additional USD conversion columns (usd_pledged_real and usd_goal_real).

To keep the model clean and avoid unnecessary duplication, the second option was chosen. The final dataset used for analysis is therefore the 2018 file only.

Step 6: Final Load

The cleaned and structured 2018 file was loaded into the Power BI data model. The original 2016 file and the appended table were disabled from loading, ensuring that the model contains only unique, non-duplicated records with the most complete set of columns.

Data Modeling

Since the dataset consists of a single fact table, there are no complex relationships to define. The modeling phase focused on two tasks: creating a dedicated Date table to enable time intelligence, and writing DAX measures and calculated columns for the analysis.

Create a Date Table

A dedicated Date table allows the use of time intelligence functions in DAX and enables drill-down filtering in visuals.

Steps:

1. In Power BI Desktop, go to the Modeling tab → New Table.
2. Enter the following DAX:

```
Date = CALENDAR(MIN('AllData'[launched]), MAX('AllData'[deadline]))
```

This creates a continuous range of dates from the earliest launch to the latest deadline.

3. Add the following calculated columns to the Date table:

```
Year          = YEAR([Date])
Month         = MONTH([Date])
Month Name    = FORMAT([Date], "MMMM")
Quarter       = QUARTER([Date])
Day of Week   = WEEKDAY([Date])
Day of Week Name = FORMAT([Date], "dddd")
```

4. Mark the table as a date table: right-click the table in the Fields pane → Mark as date table → select the Date column.

Establish Relationships

A relationship is required between the Date table and the main table so that date filters propagate to the project data.

- The main table has two date columns: launched (start of campaign) and deadline (end).
- An active relationship is created from Date[Date] to AllData[launched] (One-to-Many).
- If analysis by deadline is needed later, USERELATIONSHIP() can be used in DAX to activate the inactive relationship.

Steps:

1. Go to Model view.
2. Drag from the Date column in the Date table to the launched column in the main table.
3. Ensure Cardinality is One-to-Many and Cross-filter direction is Single.

Note: A known UI bug in Power BI Desktop may hide the OK button in the relationship dialog. The workaround is to press Tab until the button is focused, then press Enter to confirm.

Calculated Columns

The following calculated columns were added to the main table to support the analysis:

Goal Category

Groups projects by their funding goal size to allow comparison of success rates across goal ranges.

```
Goal Category =
    IF('AllData'[usd_goal_real] < 10000, "Small",
    IF('AllData'[usd_goal_real] < 50000, "Medium",
    IF('AllData'[usd_goal_real] < 100000, "Large", "Very Large")))
```

Duration Category

Groups campaigns by their length in days.

```
Duration Category =
    IF('AllData'[Duration] < 15, "Short",
    IF('AllData'[Duration] <= 30, "Medium",
    IF('AllData'[Duration] <= 60, "Long", "Very Long")))
```

Pledged vs Goal

Classifies each project based on whether it reached its funding goal.

```
Pledged vs Goal =
    IF('AllData'[usd_pledged_real] >= 'AllData'[usd_goal_real],
       "Met Goal", "Did Not Meet Goal")
```

Core DAX Measures

Measures are calculations performed dynamically at query time. The following measures were created:

Total Projects

```
Total Projects = COUNT(AllData[ID])
```

Successful Projects

```
Successful Projects = CALCULATE([Total Projects], 'AllData'[state] =
"successful")
```

Failed Projects

```
Failed Projects = CALCULATE([Total Projects], 'AllData'[state] = "failed")
```

Success Rate

```
Success Rate = DIVIDE([Successful Projects], [Total Projects])
```

Total Pledged (USD)

```
Total Pledged USD = SUM('AllData'[usd_pledged_real])
```

Average Pledged per Project

```
Avg Pledged USD = AVERAGE('AllData'[usd_pledged_real])
```

Total Goal (USD)

```
Total Goal USD = SUM('AllData'[usd_goal_real])
```

Pledged vs Goal Ratio (average across projects)

```
Avg Pledged/Goal = AVERAGEX(
    FILTER('AllData', AllData[usd_goal_real] > 0),
    DIVIDE('AllData'[usd_pledged_real], 'AllData'[usd_goal_real]))
```

Average Duration (days)

```
Avg Duration = AVERAGE('AllData'[Duration])
```

Total Backers

```
Total Backers = SUM('AllData'[backers])
```

Average Backers per Project

```
Avg Backers per Project = AVERAGE('AllData'[backers])
```

Visualisation & Reporting

The goal of this phase is to transform the cleaned data model into a visual story. The report is structured across three pages, each addressing a specific set of questions. Slicers are used throughout to make the report interactive.

Key Questions

Before building visuals, the following questions were defined to guide the report:

1. What is the overall success rate?
2. Which categories have the highest success rate?
3. How does the funding goal relate to success?

4. Does the campaign duration affect success?
5. Are there yearly trends in the number of projects?
6. Which countries have the most projects and highest success rates?
7. What is the relationship between number of backers and project success?
8. How many projects met their funding goal?

Page 1 — Overview

Purpose: provide a high-level summary that answers the most important questions within 30 seconds.

Visuals:

- Four Card visuals: Total Projects, Success Rate (formatted as %), Total Pledged USD, Total Backers.
- Clustered Bar Chart — Success Rate by main_category. Axis: main_category. Value: Success Rate. Sorted descending.
- Map — Total Projects by country. Location: country. Size: Total Projects. Tooltip: Success Rate and Total Pledged USD.
- Line Chart — Total Projects over time. Axis: Year from the Date table. Value: Total Projects. This chart shows the growth and decline of Kickstarter activity from 2009 to 2018.
- Clustered Bar + Line Chart — Total Projects and Total Backers by state. Shows the relationship between the number of backers and project outcome.
- Slicers: Year (range slider), State, Main Category.

Note: Kickstarter is an American platform founded in New York in 2009. It remained exclusive to US residents for a long time before gradually opening up to other countries. This explains the significant bias towards US projects visible on the map.

Page 2 — Category & Geography

Purpose: explore how project volume, funding, and success vary across categories and countries.

Visuals:

- Stacked Bar + Line Chart — Total Projects (stacked by state) and Total Backers (line) by main_category. Shows both volume and backer engagement per category.
- Stacked Bar Chart — Total Pledged USD (stacked by state) by main_category. Highlights which categories attract the most funding.
- Bar Chart — Successful Projects by main_category. Sorted descending.
- Bar Chart — Successful Projects by country. A logarithmic scale is applied on the Y-axis to reduce the visual dominance of the US and make other countries visible.

Note: A label “Use To Filter All Charts” was added above the country chart to guide users to click on a country and apply cross-filtering across all visuals on the page.

Page 3 — Success Factors

Purpose: analyse the factors that influence whether a project succeeds or fails.

Visuals:

- Stacked Bar + Line Chart — Total Projects (stacked by state) and Total Backers (line) by Goal Category. Reveals how the size of the funding goal relates to project outcomes.
- Stacked Bar + Line Chart — Total Projects (stacked by state) and Total Backers (line) by Duration Category. Reveals whether campaign length influences success.
- Donut Chart — Success Rate by Goal Category. Shows that Small goal projects have the highest success rate (41%), while Very Large goal projects have the lowest (5%).
- Pie Chart — Success Rate by Duration Category. Shows that campaign duration has minimal impact on success — rates range only from 23% to 31% across all duration groups.

- Stacked Bar Chart — Total Projects split by Pledged vs Goal column, broken down by state. Shows how many projects actually met their funding goal within each outcome category.

Key Insights

General

- Total projects analysed: 379K.
- Overall success rate: 35%.
- Total funding raised: 3.43bn USD.
- Total backers across all projects: 40M.
- Project volume peaked in 2015 at approximately 80K projects per year, then declined steadily through 2018.

Categories

- Dance and Theater have the highest success rates among all main categories.
- Film & Video and Music have the largest number of submitted projects.
- Games and Design attract the highest total pledged amounts.

Success Factors

- Funding goal size is the strongest predictor of success: Small goal projects (<$10K) succeed at 47%, while Very Large goal projects (>$100K) succeed at only 7%.
- Campaign duration has minimal impact on success: success rates across all duration groups range narrowly from 23% to 31%.
- Projects with more backers are strongly associated with successful outcomes, as shown by the backer distribution across project states.

Geography

- The US dominates the dataset due to Kickstarter's American origins and its phased international expansion.
- GB and CA are the second and third most active countries respectively.
- European participation increased notably from 2015 onwards.